

Practical Session 6 - Approximation Algorithms

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25/05/2026

Problem 1

In the Knapsack problem, we are given a set $A = \{a_1, \dots, a_n\}$ of items, where each a_i has a specified positive integer size s_i and a specified positive integer value v_i . We are also given a positive integer knapsack capacity B . Assume that $s_i \leq B$ for each i . The problem is to find a subset of A whose total size is at most B and for which the total value is maximized. In this problem, we will consider approximation algorithms to solve the Knapsack problem. For any subset S of A , we write s_S for the total of all the sizes in S and v_S for the total of all the values in S . Let OPT denote an optimal solution to the problem.

- (i) Consider the following greedy algorithm Alg1 to solve the Knapsack problem:
Order all the items a_i in non-increasing order of their density, which is the ratio of value to size, $\frac{v_i}{s_i}$. Make a single pass through the list, from highest to lowest density. For each item encountered, if it still fits, include it, otherwise exclude it. Prove that algorithm Alg1 does not guarantee any constant approximation ratio. That is, for any positive integer k , there is an input to the algorithm for which the total value of the set of items returned by the algorithm is at most $\frac{v_{\text{OPT}}}{k}$.
- (ii) Consider the *fractional* Knapsack problem, where we are allowed to take a fraction $0 \leq p_i \leq 1$ of item i of size $p_i s_i$ getting value $p_i v_i$. Let i be the smallest index in the ordered list (by non-increasing order of densities) such that the total size of the first i items exceeds B . Show that the greedy algorithm that adds the complete items $a_1, \dots, a_{i-1}$ and a fraction of a_i such that the sum of the sizes is B , is optimal.
- (iii) Consider the following algorithm Alg2.
If the total size of all the items is $\leq B$, then include all the items. If not, then order all the items in non-increasing order of their densities. Without loss of generality, assume that this ordering is the same as the ordering of the item indices. Find the smallest index i in the ordered list such that the total size of the first i items exceeds B (i.e., $\sum_{j=1}^i s_j > B$, but $\sum_{j=1}^{i-1} s_j \leq B$). If $v_i > \sum_{j=1}^{i-1} v_j$, then return $\{a_i\}$. Otherwise, return $\{a_1, \dots, a_{i-1}\}$. Prove that Alg2 always yields a 2-approximation to the optimal solution.

Problem 2

The Sonet Ring Loading Problem is a classical problem in communication networks. We are given a ring of n nodes numbered from 0 to $n-1$ clockwise around the ring and a set C of calls. Each call is a pair $(i, j) \in C$ originating from node i and ending at node j . We can route each call clockwise or counterclockwise around the ring. The load L_i of the link $(i, (i+1) \bmod n)$ is the number of calls routed through that link. The problem is to choose how to route the calls as to minimize $\max_{i=0, \dots, n-1} L_i$. Provide a 2-approximation algorithm for the problem.

Problem 3

In the **Max k-Cut** problem we are given an undirected graph $G = (V, E)$ and non-negative weights $w_{ij} \geq 0$ for all $(i, j) \in E$. The goal is to partition the set into k parts so as to maximize the weight of all edges whose endpoints are in different parts. Give a randomized $\frac{k-1}{k}$ -approximation algorithm for the problem.